

Assignment No. 3

Title

Uploading Sensor Data to a Web Server Using ESP8266 NodeMCU

Aim

To interface a sensor with the ESP8266 NodeMCU and upload real-time sensor data to a web server using Wi-Fi.

Theory (Students should prepare the following write-up)

Students should study and write brief notes on the following topics:

1. Introduction to IoT Data Acquisition.
2. Sensors used in IoT applications.
3. Working principle of the selected sensor (e.g., DHT11/DHT22, LM35, LDR, Soil Moisture Sensor).
4. ESP8266 NodeMCU as an IoT Gateway.
5. Client-Server Architecture.
6. HTTP Protocol and Request Methods (GET and POST).
7. Web Server Architecture.
8. Data Logging in IoT.
9. JSON Data Format.
10. REST API Basics.
11. Applications of Remote Monitoring Systems.

Equipment/Software Required

Hardware

- **ESP8266 NodeMCU**
- **DHT11/DHT22 (or any suitable IoT sensor)**
- **Breadboard**
- **Connecting Wires**
- **USB Data Cable**
- **Wi-Fi Internet Connection**

Software

- **Arduino IDE**
 - **ESP8266 Board Package**
 - **Required Sensor Library**
 - **ESP8266WiFi Library**
 - **HTTPClient Library**
 - **Web Server (Apache/XAMPP/Localhost or Remote Web Server)**
 - **PHP Script/API (for receiving sensor data)**
 - **MySQL Database (optional)**
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Procedure

Part A: Hardware Setup

1. **Connect the sensor to the ESP8266 NodeMCU.**
2. **Verify wiring connections.**
3. **Connect ESP8266 to the computer.**

Part B: Software Configuration

- 1. Install the required sensor library.**
- 2. Configure Wi-Fi credentials.**
- 3. Configure the Web Server URL/API.**
- 4. Read sensor values.**
- 5. Format the data for transmission.**
- 6. Send sensor data using an HTTP GET or POST request.**
- 7. Upload the program to the ESP8266.**

Part C: Verification

- 1. Open the Serial Monitor.**
 - 2. Verify Wi-Fi connection.**
 - 3. Observe sensor readings.**
 - 4. Confirm successful HTTP response from the server.**
 - 5. Verify that the uploaded data is displayed or stored on the server/database.**
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Observations

Students should record:

- Sensor used**
- Wi-Fi SSID**
- Server URL**
- HTTP Method (GET/POST)**
- Sensor values transmitted**
- Server response code**
- Upload interval**
- Successful data upload status**

Result

Sensor data was successfully acquired using the ESP8266 NodeMCU and uploaded to the web server over a Wi-Fi network.

Conclusion (To be written by students)

Students should write a conclusion covering:

- **Working of sensor-based IoT systems.**
 - **Steps involved in transmitting data to a web server.**
 - **Importance of HTTP communication in IoT.**
 - **Applications of web-based monitoring systems.**
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Viva-Voce Questions

- 1. What is a Web Server?**
- 2. Explain HTTP GET and POST methods.**
- 3. What is REST API?**
- 4. Why is JSON used in IoT?**
- 5. What is the function of HTTPClient Library?**
- 6. What are the advantages of web-based monitoring?**
- 7. Explain client-server architecture.**
- 8. What is data logging?**
- 9. How does ESP8266 communicate with the server?**
- 10. List applications of sensor data monitoring.**

Precautions

- **Verify Wi-Fi connectivity before uploading.**
 - **Check API URL carefully.**
 - **Ensure sensor calibration.**
 - **Verify server response.**
 - **Use stable power supply.**
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Outcome

Students will be able to interface sensors with ESP8266 NodeMCU, collect real-time sensor data, communicate with a web server using HTTP, and implement basic IoT-based remote monitoring applications.
